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Search for New Physics in anomalous jets using jet substructures with the ATLAS detector and LHC Run 3 data

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Introduction

It was the 4 July 2012 when Joseph Incandela, CMS spokesperson, affirmed: "*They combine to give us a combined significance of five standard deviations*". With these exact words, he announced, confirmed by ATLAS spokesperson Fabiola Gianotti a few minutes later, the first observation of the Higgs boson [1, 2], which completed the Standard Model (SM) framework. Fourteen years later, no comparable discovery has followed within the SM paradigm at collider experiments, strengthening the hypothesis that it is the most complete and well-tested description of nature.
In reality, SM probably is not as complete as it appears. In fact, there are many phenomena not included in the framework but confirmed by astrophysical and cosmological observations: among these we can find the neutrino mass measurement, Dark Matter [3] and baryon asymmetry. All these processes fall in the category of New Physics (NP) Beyond the Standard Model (BSM). They have been hunted by ATLAS and CMS experiments using traditional data analysis techniques, targeting specific theoretical models but, unfortunately, nothing has been observed yet and only upper limits on production cross-sections were set. There are many reasons for this unexpected outcome: wrong Physical model for experiments, wrong triggering while selecting data, or deviations too small to be observed through normal techniques.
A possible solution to some of these issues could be Anomaly Detection (AD), that is the search for data deviating from what is already known. In our field of interest, Particle Physics, we apply this method to look for anomalies (signal data), deviating from well known reacting and decaying particles (background data). The strength of this method lies in its model agnostic nature, meaning that it does not rely on a specific theoretical model to analyse data. This allows us to explore a more vast range of events, increasing the background rejection efficiency, giving us a first hint of the region where to look more in deep for New Physics with precise data analysis techniques. In this process, unsupervised Machine Learning (ML) helps speed up the process.
This thesis investigates whether such an unsupervised approach can be applied to a concrete search. Using ATLAS official Monte Carlo (MC) samples simulating the ATLAS detector during Run-3, a search for BSM resonant signals decaying into hadronic final states with large-radius jets (used to probe boosted resonance topologies) has been conducted, preparing the needed tools to finalise the analysis on real ATLAS data. Leveraging the Anomaly Score (AS) distributions computed by a Graph Neural Network (GNN) for each jet, the analysis strategy was defined, starting with the definition of Working Points (WPs), thresholds on the AS distribution corresponding to a given background rejection fraction. A crucial part of

the strategy was determining the correct modelling function for the background and signals di-jet invariant mass (m_{JJ}) distribution. After tuning and validating a Double-Sided Crystal Ball (DSCB) function to model the signals and a 5-parameter function for the background distribution, fits to the MC samples were executed, deriving the expected upper limits at 95% Confidence Level on m_{JJ} using the CLs method, at different AS WPs. The study on MC samples was concluded and fully validated by performing sensitivity studies based on a binned maximum likelihood fit to the di-jet invariant mass distribution.
The thesis is structured as follows. In Chapter 1 the Standard Model is presented, together with theories of Physics beyond it. The LHC collider and the ATLAS experiment are discussed in Chapter 2. Chapter 3 presents the Anomaly Detection approach. The typical workflow of a Particle Physics analysis is described in Chapter 4. Chapter 5 presents the analysis conducted during this study and its results. Finally, conclusions are drawn in Chapter 6.

First index draft:

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Chapter 1

(Beyond) The Standard Model

The Standard Model (SM) of Particle Physics is a relativistic Quantum Field Theory (QFT) describing three out of four fundamental interactions, strong, weak and electromagnetic (EM), and the behavior of elementray particles under them. For a long time it was attempted to merge the fourth fundamental interaction, gravity, with the SM framework, but those attempts have not been validated yet by experimental observations. Nevertheless, for our purposes, gravity does not play a significant role since the energy scales involved in the processes at a collider experiment, as the LHC, are much lower than the Planck scale, where gravitational effects become relevant.
What is remarkable about the SM is that it is a very well tested theory, with predictions confirmed by experimental observations with a very high precision. The discovery of the Higgs boson in 2012 at the LHC, which completed the SM framework, is a clear example of this.
Despite the success of the SM, there are many phenomena that are not explained by it, but are confirmed by astrophysical and cosmological observations, opening the road for Physics Beyond the Standard Model.
This chapter presentes a brief overview of the SM, highlighting its major features. In addition, hints for Physics Beyond it are exposed.

1.1 An overview of the Standard Model

The fundamental constituents of Universe are precisely described by the SM. They are called elementary particles, which are mathematically described as quantum fields $\psi(x)$ and their dynamics is modelled by Lagrangian densities $\mathcal{L}(\psi, \partial_\mu \psi)$ [4].
Such particles can be graphically pictured as presented in Fig. 1.1. They are divided into two fundamental classes based on their spin quantum number. The first class is composed by *fermions*, which cover the first three column of Fig. 1.1 and are organised in three *generations* of increasing proper mass; they have spin 1/2 and constitute *matter*. The second class is those of *bosons*, with integer spin, which *mediate* fundamental interactions and generate mass.
Fermions are further classified into two distinct categories depending on their participation in the strong interaction. The first category is relative to *quarks*. They carry *colour charge* (red, green, blue), making them triplets under $SU(3)$ and therefore interact via all fundamental forces, among which the strong one. Quarks

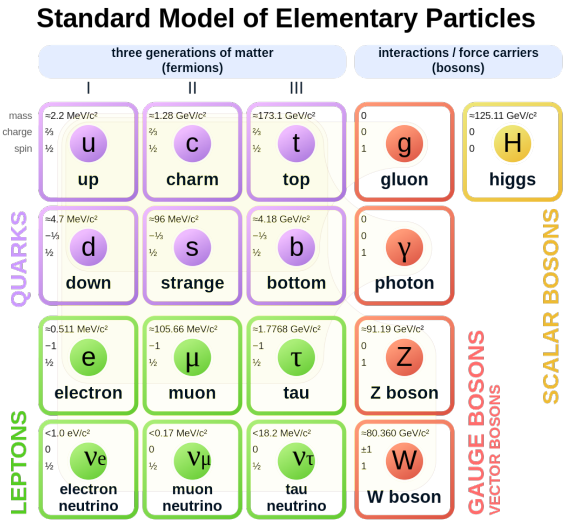


Figure 1.1 The SM framework. Figure from Ref. [5].

exist in six flavors, organized into three generations of weak SU(2) doublets:

$$\begin{pmatrix} u \\ d \end{pmatrix}, \quad \begin{pmatrix} c \\ s \end{pmatrix}, \quad \begin{pmatrix} t \\ b \end{pmatrix} \tag{1.1}$$

Quarks carry a fractional electromagnetic charge: the *up-type* ones (u, c, t) have a charge of $+2/3$, while the *down-type* ones (d, s, b) carry a $-1/3$ charge.
The second category of fermions is associated to *leptons*; they are six and similarly grouped into three generations:

$$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix}, \quad \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix}, \quad \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix} \tag{1.2}$$

The charged leptons (e^-, μ^-, τ^-) carry an electric charge of -1 and interact via electromagnetic and weak forces. Neutrinos $(\nu_e, \nu_\mu, \nu_\tau)$ are electrically neutral and interact exclusively via the weak force.
Both quarks and charged leptons, have their own anti-particles, which have the same mass of the matter components but opposite quantum numbers. Regarding neutrinos, it is still debated and to be proved whether they have a distinct anti-particles or they are Majorana fermions, being their own anti-particles.
The second class of particles is associated to mediatiors and mass. Spin 1 vector bosons, known as *gauge bosons*, are the mediators of three of the fundamental interactions, which arise naturally from the requirement of local gauge invariance. Among those fields we find the 8 massless gluons g mediating the strong interaction, the massless photon γ is associated to electromagnetic interaction and 3 massive vector bosons, $W^\pm$ and Z^0, with masses $m_W = 80.36$ GeV, $m_Z = 91.19$ GeV, mediating the weak interaction.
In conclusion, a spin 0 scalar particle, the *Higgs boson* H, with a mass $m_H = 125.13$ GeV, completes the particle spectrum, arising as a physical state from the

spontaneous breaking of electroweak symmetry.

The SM is a gauge theory, meaning that the Lagrangian is invariant under local transformations of the fields. The Lagrangian density describing the SM dynamics can be written as:

$$\mathcal{L}_{\text{SM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - i\bar{\psi}\gamma_\mu D^\mu\psi + (D_\mu\phi)^\dagger(D^\mu\phi) - V(\phi) + \bar{\psi}_i y_{ij}\psi_j\phi + h.c., \quad (1.3)$$

The gauge group of the SM, under which such lagrangian is invariant, is a non-abelian symmetry group defined as:

$$SU(3)_C \otimes SU(2)_L \otimes U(1)_Y \quad (1.4)$$

where $SU(3)_C$ describes the strong interaction via Quantum Chromodynamics (QCD) with subscript C standing for colour, and $SU(2)_L \otimes U(1)_Y$ represents the unified electroweak (EW) interaction, with L associated to the left component of fermions and Y to hypercharge.
The terms of the equation 1.3 above are presented in detail in the following sections.

1.1.1 Quantum Electrodynamics

The first and simplest gauge theory within the SM framework is Quantum Electrodynamics (QED). It describes the electromagnetic interactions between a charged massive fermion, with mass m and represented by a field $\psi(x)$, and the photon, the massless gauge boson represented by the vector field A_μ. Using the Lagrangian formalism [6], QED is constructed starting from the free Dirac Lagrangian for a massive fermion [7]:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi \quad (1.5)$$

In the equation above, $\bar{\psi} = \psi^\dagger\gamma^0$ is the Dirac adjoint of the field ψ and γ^μ are the Dirac matrices derived from Pauli matrices σ_i (i =1,2,3 and μ=0,1,2,3):

$$\gamma^0 = \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \quad (1.6)$$

The Dirac Lagrangian 1.5 is invariant under a global $U(1)$ phase transformation $\psi(x) \rightarrow e^{ie\alpha}\psi(x)$, with e the electric charge of the fermion and α a constant phase. Such invariance is not preserved under a local $U(1)$ phase transformation $\psi(x) \rightarrow e^{ie\alpha(x)}\psi(x)$, where $\alpha(x)$ is a space-time dependent phase. In order to preserve the invariance, the partial derivative ∂_μ is replaced by the covariant derivative D_μ:

$$D_\mu = \partial_\mu + ieA_\mu(x) \quad (1.7)$$

where $A_\mu(x)$ is a vector gauge field representing the photon field, which transforms as $A_\mu(x) \rightarrow A_\mu(x) - \partial_\mu\alpha(x)$. Including the gauge-invariant kinetic term to take into account for the new field A_μ, we derive the full QED Lagrangian:

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad (1.8)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor. The interaction term $-e\bar{\psi}\gamma^\mu\psi A_\mu$ rules the coupling of charged fermions to photons with a strength proportional to the electric charge e.
The invariance under a local $U(1)$ transformation gurantees, via the Noether's theorem [8], the existence of a conserved current and charge which, in the case of QED, are the electromagnetic current and the electric charge, respectively.

1.1.2 Quantum Chromodynamics

The building of the Lagrangian for QCD is similar to the one for QED: the Dirac Lagrangian is required to be invariant under a local symmetry group, which in this case is the non-Abelian $SU(3)_C$ group, where C represents the triplet color degree of freedom (red, green, blue), the conserved charge of QCD [9]. This theory models the strong interaction between *quarks*, fundamental constituents of hadrons represented as fermionic fields $q(x)$, and *gluons*, the massless gauge bosons represented as vector fields $G^a_{\mu\nu}(x)$ with $a = 1, ..., 8$. Similarly to QED, the local transformation of the quark field is defined as $q(x) \to q'(x) = e^{ig_s T^a \alpha_a(x)} q(x)$, with g_s the strong coupling constant and $\alpha_a(x)$ the local phase. The term T^a appearing in the previous transformation equation is associated to the generators of the $SU(3)_C$ group, which can be expressed via the Gell-Mann matrices as $T^a = \frac{1}{2}\lambda^a$; such generators satisfy $[T^a, T^a] = if^{abc}T^c$ with f^{abc} being the structure constants of $SU(3)$.
Also in this case, a covariant derivative is introduced to preserve the invariance of the Lagrangian under local transformations:

$$D_\mu = \partial_\mu + ig_s T_a G^a_\mu \tag{1.9}$$

The non-Abelian structure of the group, gives rise to the QCD field tensor:

$$F^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu - g_s f^{abc} G^b_\mu G^c_\nu \tag{1.10}$$

The presence of the non-linear structure factor f^{abc} both in the commutators and in the tensor, is associated to a particular feature of QCD: the triple and quadruple gluon self-interaction of gluons, a feature not present in QED.
Finally, the Lagrangian describing the dynamics of QCD is written as:

$$\mathcal{L}_{\text{QCD}} = \bar{q}\left(i\gamma^\mu D_\mu - m\right) q - \frac{1}{4} F^a_{\mu\nu} F^{a,\mu\nu} \tag{1.11}$$

The term g_s, cited before in the local transformation, is usually presented as $\alpha_s = \frac{g_s^2}{4\pi}$, which is the running coupling constant of QCD, whose value depends on the energy scale Q^2 of the process. The non-Abelian nature of QCD gives rise to a peculiar behavior of $\alpha_s(Q^2)$. In fact, at high energy scales (meaning short distances), $\alpha_s(Q^2)$ decreases logrithmically, giving rise to the *asymptotic freedom*, enabling accurate calculations via perturbative QCD (pQCD). On the other hand, at low energy scales (meaning long distances), $\alpha_s(Q^2)$ increases, giving rise to the *color confinement*, which prevents colored particles from existing as isolated free states. In this regime, quarks and gluons undergo fragmentation and hadronization, producing collimated sprays of energetic hadrons, known as *jets*.

1.1.3 Electroweak model

Making use of the Glashow-Weinberg-Salam (GWS) model [7, 10], it possible to derive a unified description of *electromagnetic* interaction, already presented in Sec. 1.1.1, and the final third interaction, the *weak* one.
To derive the Electroweak (EW) Lagrangian, the steps to follow are similar to those already presented for QED and QCD. In the first place, the symmetry group under which the Lagrangian is invariant is defined, which is $SU(2)_L \otimes U(1)_Y$ in this case, with Y known as hypercharge, the quantum number of EW interactions.
Secondly, the description of fermions and gauge bosons is defined. Concerning the latter, we have four bosons: W_μ^i with $i = 1, 2, 3$ associated to $SU(2)_L$ and B_μ associated to $U(1)_Y$, with coupling constants g and g' respectively. Moving to fermions, they are chirality eigenstates. This means that they can be expressed as left-handed and right-handed components, which are defined as $\psi_{L,R} = \frac{1}{2}(1 \mp \gamma^5)\psi$, with $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. The left-handed components of fermions transform as doublets under $SU(2)_L$ and are described as $L = (\nu, l^-)$ for leptons and $Q = (u, d)$ for quarks. They have quantum numbers $(T = 1/2, T_3 = \pm 1/2)$, with T being the isospin. The right-handed components, instead, transform as singlets under $U(1)_Y$, with quantum numbers $(T = 0, T_3 = 0)$.
The transformation of the fermion components is defined in a compact way as:

$$\begin{aligned} \psi_L(x) &\longrightarrow \psi'_L(x) = e^{i\vec{\alpha}(x)\cdot\vec{\tau} + i\beta(x)Y}\psi_L(x), \\ \psi_R(x) &\longrightarrow \psi'_R(x) = e^{i\beta(x)Y}\psi_R(x) \end{aligned} \tag{1.12}$$

where $\vec{\tau} = \frac{1}{2}\vec{\sigma}$ are the weak isospin, generators of $SU(2)_L$.
Electric charge Q, weak isospin T_3, and weak hypercharge Y are connected by the relation:

$$Q = T_3 + \frac{Y}{2} \tag{1.13}$$

The following step towards the EW Lagrangian, is to introduce the covariant derivative to preserve the invariance of the Lagrangian under local transformations. The derivative is defined as:

$$D_\mu = \partial_\mu + ig\frac{\tau^a}{2}W_\mu^a(x) + ig'\frac{Y}{2}B_\mu(x) \tag{1.14}$$

At this point, the EW Lagrangian can be written as:

$$\mathcal{L}_{\text{EW}} = -\frac{1}{4}W^{a,\mu\nu}W^a_{\mu\nu} - \frac{1}{4}B^{\mu\nu}B_{\mu\nu} + i\bar{\psi}\gamma^\mu D_\mu\psi \tag{1.15}$$

where $W^{a,\mu\nu}$ and $B^{\mu\nu}$ are the field strength tensors associated to the gauge bosons W_μ^a and B_μ, respectively.
In conclusion, if a rotaion is applied to the W_μ^a and B_μ fields, the four physical gauge bosons $\gamma, Z, W^\pm$ are obtained as:

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix}\begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix}, \quad W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2) \tag{1.16}$$

The parameter appearing in the previous equation is the Weinberg angle θ_W, which is defined as $\tan\theta_W = g'/g$.

The EW Lagrangian 1.15 does not provide a proper description about the mass of the bosons involved, which is in contrast with the experimental observations of massive $W^{\pm}$ and Z^0 bosons. The solution to this issue is provided by the Brout-Englert-Higgs mechanism (BEH), which is presented in the following section.

1.1.4 The Brout-Englert-Higgs mechanism

The BEH mechanism provides a theoretical framework to generate masses for the gauge bosons and fermions of the SM, while preserving the local gauge invariance of the Lagrangian. The mechanism is based on the concept of spontaneous symmetry breaking (SSB) of the group $SU(3)_C \otimes SU(2)_L \otimes U(1)_Y$ into $SU(2)_L \otimes U(1)_{EM}$ [11]. The SSB is realized by introducing a complex scalar field ϕ, which is a doublet under $SU(2)_L$ and carries hypercharge $Y = 1/2$. It is defined as:

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix} \tag{1.17}$$

The potential associated to the scalar field is defined as:

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \frac{\lambda}{2} \left(\phi^\dagger \phi\right)^2 \tag{1.18}$$

where μ^2 and λ are constants. Among the two constants, λ must be positive to ensure the stability of the potential, while μ^2 can be either positive or negative. The shape of the potential is determined by the sign of μ^2. If $-\mu^2 > 0$, the potential has a single minimum at $\phi = 0$, preserving the symmetry. If $-\mu^2 < 0$, the potential gets the well-known mexican hat shape presented in Fig 1.2.

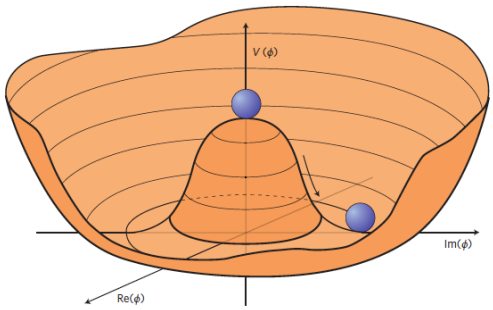


Figure 1.2 Representation of the Higgs potential $V(\phi)$. Figure from Ref. [12].

In the latter case, the vacuum state, as known as the minimum of the potential, is degenerate and associated to a non-zero value of the scalar field ϕ. In particular, the vacuum expectation value (VEV) v of the Higgs field, defined as $v = \sqrt{\mu^2/\lambda}$, is not null. As direct consqeuence, the global $U(1)_Y$ symmetry is spontaneously broken. The VEV is given by:

$$\phi^\dagger \phi = \frac{\mu^2}{2\lambda} = \frac{v^2}{2}, \quad \phi_0 = \frac{v}{\sqrt{2}} \tag{1.19}$$

Since the theory is built to be gauge invariant, in order to remove the unphysical degrees of freedom, the unitary gauge transformation is applied to the scalar field ϕ.

In this gauge, the Higgs field can be expressed as:

$$\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} \tag{1.20}$$

where $h(x)$ is the Higgs boson scalar field.
Using the definition of ϕ presented in Eq. 1.20, the EW Lagrangian density can be written as:

$$\begin{aligned} &\mathcal{L}_{\text{SM}} = -\tfrac{1}{4} W^{a,\mu\nu} W^{a}_{\mu\nu} - \tfrac{1}{4} B^{\mu\nu} B_{\mu\nu} + i\overline{\psi}\gamma^{\mu} D_{\mu}\psi + \\ &-Y^{l}_{ij}\overline{L_i}\phi\, e_j - Y^{\nu}_{ij}\overline{e_i}\phi\, L_j - Y^{d}_{ij}\overline{Q_i}\phi\, d'_j - Y^{u}_{ij}\overline{Q_i}\phi\, u_j + \text{h.c.} \\ &+(D^{\mu}\phi)^{\dagger}(D_{\mu}\phi) - V(\phi) \end{aligned} \tag{1.21}$$

where the term D_μ is the covariant derivative defined in Eq. 1.14 and Y_{ij} are the Yukawa couplings. Each line of the Lagrangian density 1.21 has as specific role. The first line represents the gauge bosons and how they interact with fermions, the second one contains the Yukawa couplings which is associated to Higgs coupling with fermions, and the third line contains the kinetic term for the Higgs field and its potential.
The Higgs potential $V(\phi)$, defined in Eq. 1.18, with che chosen gauge, can be expressed as:

$$V(h) = \frac{1}{2} m_h^2 h^2 + \frac{m_h^2}{2v} h^3 + \frac{m_h^2}{8v^2} h^4 \tag{1.22}$$

The m_h term appearing in the previous equation is the mass of the Higgs boson, which is a free parameter of the SM, depending on the parameters of the potential, as $m_h = \sqrt{2\lambda v^2} = \sqrt{2\mu^2}$. The cubic and quartic terms in h represent self-interactions of the Higgs boson.
The Higgs boson mass has been measured by the ATLAS and CMS collaborations [1, 2] to be $m_h = (125.13 \pm 0.11)$ GeV, which is consistent with the SM predictions. From this measurement, the value of VEV v can be derived as $v \approx 246$ GeV.

1.2 Hints of Physics Beyond the Standard Model

The SM presented in the previous section 1.1 is a very successful theory, which has been confirmed by experimental observations with a very high precision. Despite its success, the SM probably is not a complete theory, given the many open questions that are not aswered by the SM and motivate the search for Physics Beyond the Standard Model (BSM).
Among these questions, we find the nature of the Higgs boson, whether it is the only scalar particle or it is part of an extended Higgs sector. In fact, many extensions of the SM predict the existence of additional scalar or pseudo-scalar states, such as two-Higgs-doublet models (2HDMs), which are currently studied at the LHC by the ATLAS collaboration [13].
Together with theories to be proven, as the one just cited, there are many other proved by astrophysical and cosmological observations. The first one is the existence of Dark Matter (DM), which is known to compose 26.4% of the Universe, together

with a 68.7% of Dark Energy and a 4.9% of ordinary matter [14]. The presence of DM is inferred from its gravitational effects on visible matter, such as galaxies rotation curves [15], galaxy-galaxy gravitational lensing [16] and the Bullet Cluster [17], the result of the collision of two galaxy clusters. In order to observe DM at collider experiments, the approach, knwon as indirect search, is to detect decay products of DM particles. This type of search was not able to provide any evidence of DM particles yet, leaving it an ongoing study at the LHC.
Another open question is the matter-antimatter asymmetry in the Universe, also known as baryon asymmetry. In fact, in the early Universe there was balace between matter and antimatter, but today the Universe, after the cooling down of the Universe, the balance is slightly broken, in favour of matter. Such unbalance could be associated to a violation of the CP symmetry, which is a combination of charge conjugation C and parity P. This motivates the search for new sources of CP violation, which could be provided by BSM theories.
One final question is the hierarchy problem, which is associated to the Higgs boson mass. In fact, the Higgs boson mass value is subject to quantum loop corrections, which are quadratically divergent. This means that the Higgs boson mass receives contributions from all energies up to the Planck scale (10^{19} GeV). Since the Higgs boson mass has been measured to be 125 GeV, an unnatural fine-tuning of the parameters of the SM would be required with large cancellations between different terms in radiative correction. This motivates the search for new physics at the TeV scale, which could provide a natural explanation for the Higgs boson mass.

1.2.1 New Physics at colliders

After reading the BSM cases presented above, and many others not cited, it appears to be quite clear that New Physics Beyond the Standard Model should exist and, in principle, it should be accessible at the energy scales probed by the LHC. Given the lack of observations at CERN experiments, it is worth analysing the approach used to probe NP. This kind of experiments are conducted with two main different approaches: *precision measurements*, which are sensitive to small deviations from the SM predictions, and *direct searches*, which aim to produce new particles in high-energy collisions and probe them. The latter direction has been tradionally pursued by targeting a specific BSM model, which is known as *model-dependent* search. In this case, the search is optimized for a specific model and its parameters. The results of such searches are usually presented as exclusion limits on the space of parameters of the model.
A suitable alternative to model-dependent searches is the *model-independent* or *model-agnostic* approach, which searches for deviations from the SM predictions in a general way, without assuming any specific BSM model. This approach is more flexible and can be sensitive to a wider range of BSM scenarios. This type of search is gaining more and more interest in the last years in the LHC program for Run 3. A model-agnostic search for NP is the topic of this thesis, with the goal to prove its flexibility and sensitivity. Two possible BSM theories have been chosen as benchmark models to test the strategy, Heavy Vector Triplets (HVT) which could explain new narrow resonances, and Dark Jets (DJ) which are QCD-like jets produced by dark

sectors.

1.2.2 Heavy Vector Triplets

Popular studied BSM scenarios are those presented in Simplified Models [18], which assume that NP are associated with new narrow resonances. Such assumption has the direct consequence of reducing the number of free parameters of the model, making it more predictive and easier to test.
The simplest and most studied Simplified Model is the Heavy Vector Triplet (HVT) model [19], which predicts the existence of elctroweak charged spin 1 bosons with a behavior similar to the SM W and Z bosons. This model introduces in the SM Lagrangian the presence of the real vector field V_μ^a $(a = 1, 2, 3)$, which is a triplet under $SU(2)_W$ and has zero hypercharge. Making a combination of these components, the physical states of the HVT model are obtained as:

$$V_\mu^\pm = \frac{V_\mu^1 \mp iV_\mu^2}{\sqrt{2}}, \quad V_\mu^0 = V_\mu^3 \tag{1.23}$$

The HVT undergo Drell-Yan decays with highest branching ratios with two leptons, two quarks or two bosons in the final state.
ATLAS and CMS collaborations have conducted many analyses to search for HVT decaying mainly in fully hadronic final states, which are the most abundant ones. The results of these searches are usually presented as exclusion limits at 95% confidence level on the mass of the HVT bosons, which are currently excluded in the range $[1.3, 3.1]$ TeV by ATLAS [20] and $[3.3, 3.5]$ TeV by CMS [21]. A typical plot of these kind of searches can be found in Fig. 1.3, showing the result of the ATLAS analysis.

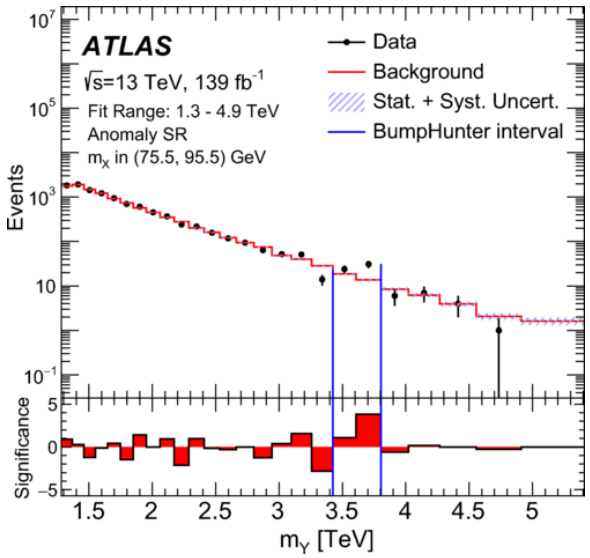


Figure 1.3 BumpHunter scan in the Y boson invariant mass distribution within the HVT framework. Figure from Ref. [20].

1.2.3 Dark Jets

The dark QCD model, a strongly coupled dark sector, has gained interest in the last years as a possible explanation for DM. [22] In this model, the dark sector is composed of dark quarks and gluons, which are charged under a new gauge group $SU(N)_D$ and interact via a new strong force, similar to QCD.
The dark quarks can be pair-produced at colliders via a mediator particle. Once produced, they undergo a shower and hadronization process, similar to QCD, producing collimated sprays of dark hadrons, known as dark jets. This process is predicted to happen at a scale Λ_D, which is the confinement scale of the dark sector, with the assumption that $\Lambda_D << \sqrt{s}$, where $\sqrt{s}$ is the center-of-mass energy of the collision.
The commonly known portal is the mediator particle involved in the interaction between the SM and the dark sector. The experimental signature of those processes at colliders will be jets with missing energy or general features not compatible with QCD.
ATLAS and CMS collaborations have conducted many analyses to search for dark jets. In particular, ATLAS has conducted a search for a $Z' \rightarrow q_D\overline{q}_D$ resonance [23], where q_D is a dark quark and Z' is probed in the range of $[1.5, 3.5]$ TeV. The observed limit on the mass for such new resonance has been set up to 3.5 TeV, as presented in Fig. 1.4.

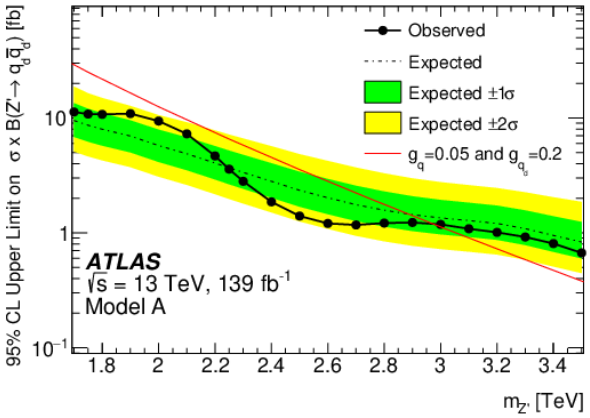


Figure 1.4 Upper limits at 95% CL on the cross section times the branching ratio for a dark QCD process. Figure from Ref. [20].

Chapter 2

The LHC and the ATLAS experiments

The history of the European Organization for Nuclear Research (CERN) traces back to the post-WWII era, when European physicists, including Edoardo Amaldi, Niels Bohr and Pierre Auger, promoted a collaborative international laboratory dedicated to fundamental science. CERN was established in September 1954 and today it hosts the world's largest and most energetic accelerator complex, centered around the Large Hadron Collider (LHC).
In the following, after a description of the LHC machine, the discussion covers the architecture of the ATLAS (A Toroidal LHC ApparatuS) detector, one of the four major experiments at CERN, including its main sub-systems and simulation procedures. A special focus is given to the proton-proton collision dataset collected by ATLAS at a center-of-mass energy of 13.6 TeV during LHC Run 3, which forms the basis for this study.

2.1 The LHC

The Large Hadron Collider (LHC) at CERN, with its 27 kilometres ring [25], is the world largest particle accelerator. It has four major interaction points which host the major CERN's experiments: ATLAS, CMS, LHCb and ALICE. LHC is situated around 100 metres underground to shield from radiation and for both economic[1]and geological reasons. With its first run on 10 September 2008, it currently accelerates hadrons. Depending on the run, protons or lead nuclei are collided, with a center-of-mass energy of 13.6 TeV and 5.36 TeV respectively. In order to reach such high energies, a complex system of pre-accelaration is adopted, as presented in Fig. 2.1.

[1]Building LHC on the surface would have required the purchase of properties to do so and an under the open sky structure with a 27 km circumference would have been more expensive than agreeing with landowner and excavate a tunnel underground.

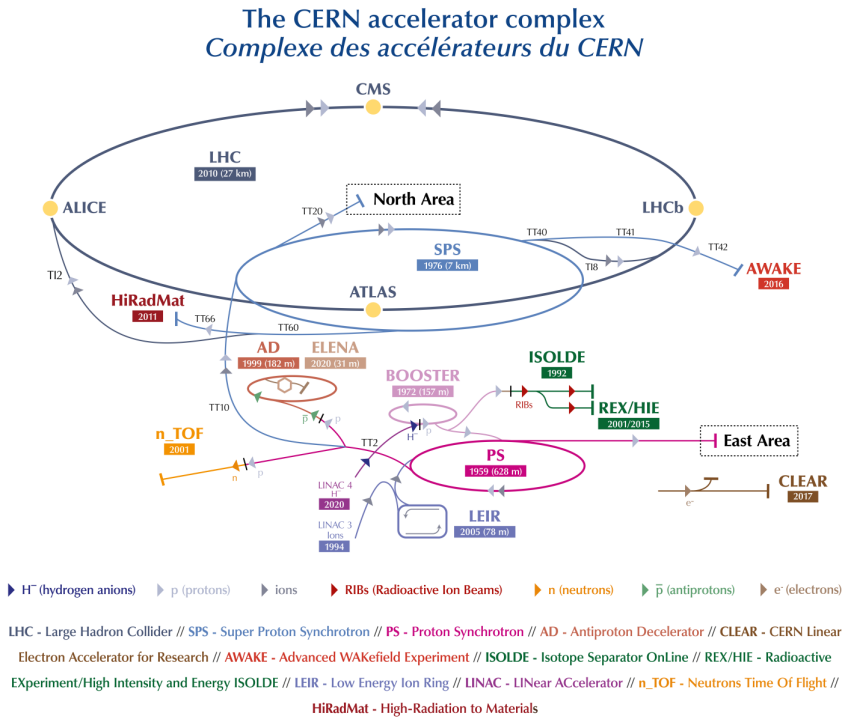


Figure 2.1 CERN accelerating system. Figure from Ref. [24].

2.1.1 CERN's accelerator system

For proton-proton (pp) collisions, the process begins with the extraction of negative hydrogen ions (H^-), which are initially accelerated to an energy of 160 MeV by the Linear Accelerator 4 (LINAC4). Upon injection into the Proton Synchrotron Booster (PSB), the ions are stripped of their two electrons leaving only the bare protons, which are grouped into bunches and boosted to an energy of 2 GeV. The protons then enter the Proton Synchrotron (PS), reaching an energy of 25 GeV. The final step takes place in the Super Proton Synchrotron (SPS) that accelerates the proton bunches up to the LHC injection energy of 450 GeV. Finally, the proton bunches are injected into the two counter-rotating beam pipes of the LHC, where they are accelerated to their maximum operational energy of 6.8 TeV per beam.
In heavy-ion operations (primarily based on lead nuclei, ^{208}Pb), ions are initially accelerated by the Linear Accelerator 3 (LINAC3) to an energy of 4.2 MeV per nucleon. The nucleons are then accumulated and boosted to 72 MeV per nucleon inside the Low Energy Ion Ring (LEIR). In the next stage, the PS accelerates ions to 5.9 GeV per nucleon, before transferring them to the SPS where they reach an energy of 177 GeV per nucleon. When injected into the main LHC ring, the heavy-ion beams are accelerated to a final energy of 2.68 TeV per nucleon.
Particles are accelerated up to such high energies by means of radio-frequency (RF) cavities, resonant structures generating an oscillating electric field, with a fixed frequency of 400MHz, which accelerates charged particles each time they pass through. Particles with the target energy passing trough the cavity while the field is zero,

will not be accelerated, while more (less) energetic particles will be decelerated (accelerated). Using this technique, it is ensured that particles stays in the cited bunches.

2.1.2 The LHC parameters

The LHC is composed by two circular beam pipes, denominated Beam 1 and Beam 2. Inside them, two particle beams, composed by a maximum of 2808 bunches with approximately 1.15×10^{11} protons per bunch and spaced by 25 ns, travel separately close to the speed of light in opposite direction. There are eight Interaction Points (IPs) along the LHC ring, but only four of them are used for collisions, hosting the main LHC experiments: ATLAS at IP1, ALICE (A Large Ion Collider Experiment) at IP2, CMS (Compact Muon Solenoid) at IP5 and LHCb (LHC beauty) at IP8. The beam pipes and the interaction points are presented in Fig. 2.2.

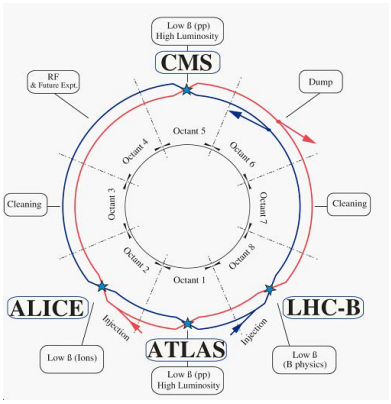


Figure 2.2 Schematic layout of the LHC (Beam 1 clockwise, Beam 2 anticlockwise). Figure from Ref. [26].

Due to the very high speed, when the two beams collide, the energy of the center of mass, known as $\sqrt{s}$, is twice the energy of each beam (which is the same). In fact, knowing that $E^2 = m^2c^4 + p^2c^2$ and using the natural unit system ([m]=eV/c^2, [p]=eV/c), we get: $E^2 = m^2 + p^2 \simeq p^2$, where at the end an ultra-relativistic particle (p»m) was assumed. This information is used in the norm of the four-momentum of two colliding particles:

$$\sqrt{s} = \sqrt{P_\mu P^\mu} = \sqrt{E_{tot}^2 - |\vec{p_{tot}}|^2} = \sqrt{(E_1 + E_2)^2 - |\vec{p_1} - \vec{p_2}|^2} \simeq 2E \qquad (2.1)$$

where we assumed $E_1 = E_2$. The 6.8 TeV of energy reached by LHC protons confirm the assumption p»m and it leads to the known value of $\sqrt{s} = 13.6$ TeV.
Accelerated beams are directed and curved by superconducting electromagnets with a magnetic field of up to 8.3 T [27]: 1232 dipole magnets and 392 quadrupole magnets are used to mantain the beams focused. Magnets are kept at a temperature of 1.9 K to ensure a 11600 A electric current which is needed for the intense magnetic field. Despite thier essential contribution, magnets also represent one of the main limitations of this apparatus. In fact, from the Lorentz force equation $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$, it is possible to get a connection between the external magnetic

field $\vec{B}$ (the one generated by superconducting magnets), the momentum p of the particle circulating inside the beams and its radius of curvature R, which is:

$$p[GeV] = 0.3B[T]R[m] \tag{2.2}$$

It is evident from Eq. 2.2 that, to keep a constant momentum p of the particles circulating inside the beams, a higher magnetic field B is required in case of reduction of the radius of curvature R. Unfortunately, current technologies set a limit to the magnetic field generated by a magnet, limiting also the minum dimension of an accelerator ring.
Performance of the LHC are evaluated based on an essential parameter: the instantaneous luminosity $\mathcal{L}$. It defines the number of collisions per cm^2 and per second produced in the collider and thus it is measured in $cm^{-2}s^{-1}$. It is computed as follows [28]:

$$\mathcal{L} = \frac{N_b^2 n_b f_{rev} \gamma}{4\pi\epsilon_n \beta^*} F \tag{2.3}$$

where: N_b is the number of particles in a bunch, n_b is the number of bunches in a beam, ϵ_n is the normalized transverse beam emittance, β^* is the beta function at the collision point, γ is the Lorentz relativistic factor computed as $\gamma = 1/\sqrt{1 - v^2/c^2}$, f_{rev} is the revolution frequency. There is a connection between the beta function and the transverse beam size σ, which is $\sigma = \sqrt{\epsilon_n \beta^*/\gamma}$. F represents the geometric luminosity reduction factor:

$$F = \frac{1}{\sqrt{1 + \left(\frac{\theta_c \sigma_z}{2\sigma^*}\right)^2}} \tag{2.4}$$

where θ_c is the full crossing angle at the interaction point, σ_z is the RMS bunch length and σ^* is the transverse RMS beam size at the interaction point. If during operations the instantaneous luminosity goes below a fixed threshold, beams are dumped and a new fill of the LHC is performed.
During LHC Run 3, $\mathcal{L}$ gets the value of 10^{34} cm$^{-2}s^{-1}$ for both ATLAS and CMS experiments, while for LHCb and ALICE it is 10^{32} cm$^{-2}s^{-1}$ and 10^{27} cm$^{-2}s^{-1}$ respectively.
The instantaneous luminosity is linked with the number of collisions per second :

$$\frac{dN}{dt} = \mathcal{L}\sigma_{tot} \tag{2.5}$$

where σ_{tot} is the total cross section, which is the measure of the probability of a specific interaction process to happen. If the instantaneous luminosity is integrated over time, the number of events collected is obtained:

$$N_{events} = \sigma_{tot} \int \mathcal{L} dt \tag{2.6}$$

2.1.3 LHC Run Schedule

The LHC operations are structured into extended physics data-taking campaigns named *Runs*, which are separated by planned technical maintenance and machine

upgrade periods, which are known as Long Shutdowns (LS). It was Run 1, from 2010 to 2013, that started the LHC physics program, operating at center-of-mass energies of $\sqrt{s} = 7$ TeV and 8 TeV, with a delivered integrated luminosity of 28 fb^{-1}. Run 1 culminated in the discovery of the Higgs boson by the ATLAS and CMS collaborations. Following upgrades during LS1, meant to enable higher-energy operations, Run 2 took place between 2015 and 2018 at $\sqrt{s} = 13$ TeV, delivering an integrated luminosity of approximately 156 fb^{-1}. After another round of upgrades during LS2 the collider entered the Run 3 era in 2022, which lasted until June 2026, operating at an increased center-of-mass energy of $\sqrt{s} = 13.6$ TeV. After having delivered an integrated luminosity of approximately 351 fb^{-1}, the LHC was turned off for the begin of LS3. This new stop will be used to prepare the LHC for the High-Luminosity LHC (HL-LHC) era, which is expected to start in 2029 and will last until 2042. The HL-LHC will operate at the same center-of-mass energy of Run 3 and is expected to deliver an integrated luminosity of 3000 fb^{-1} during the future Runs.
A graph showing the LHC delivered integrated luminosity during the different Runs, together with the recorded luminosity by ATLAS, is shown in Fig. 2.3.

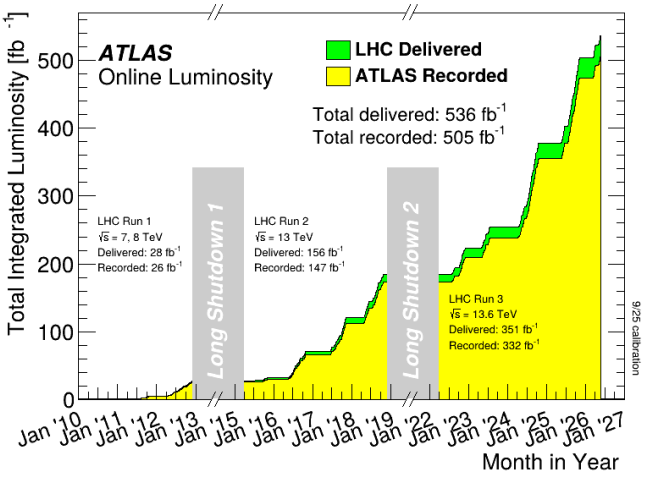


Figure 2.3 Cumulative integrated luminosity delivered to and recorded by the ATLAS Experiment during LHC Run 1, 2 and 3. Figure from Ref. [29].

2.2 The ATLAS detector

A Toroidal LHC ApparatuS (ATLAS) [30] is the world largest particle detector, situated at Interaction Point 1 at CERN. 44m in length and 25m in height are required to investigate collisions between particles accelerated by the LHC, to study phenomena such dark matter, the Higgs boson and top quark. The ATLAS detector is composed by multiple sub-detectors organized in layers with a cylindrical symmetry around the beam pipe. Among the main sub-detectors we can find: the Inner Detector (ID), the Electromagnetic Calorimeter (ECAL), the Hadronic Calorimeter (HCAL) and the Muon Spectrometer (MS), as shown in Fig. 2.4.

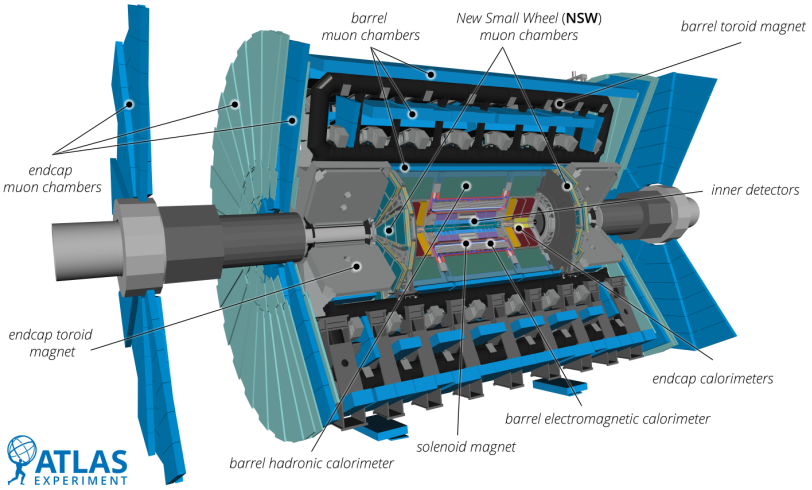


Figure 2.4 ATLAS schematic. Figure form Ref. [31].

2.2.1 Coordinate system

Before analysing the ATLAS detector, it is important to introduce the coordinate system used to describe the position of particles and their momenta inside the sub-detectors. In figure 2.5 an overview of the system is presented.

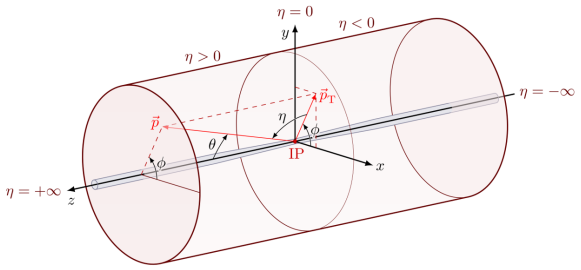


Figure 2.5 Schematic of ATLAS coordinate system

Cartesian axes (x,y,z) are constructed as follows. The origin is set in the interaction point (IP). From it, the z-axis departs parallel to the beam tunnel with positive z-values in anticlockwise direction. As direct consequence, the x-y plane is perpendicular to it, with positive x-values towards the center of the LHC and positive y-values upward. On this Cartesian system, a more natural cylindrical coordinate system (r, θ, ϕ) is defined to describe geometric position and components of particle momenta. r is the absolute distance from the interaction point. ϕ is the azimuth angle, measured around the z-axis, with values $[-\pi, \pi]$. θ is the polar angle, measured starting from the z-axis, with values $[0, \pi]$.
Two more coordinates are defined to describe the position of particles in the detector: the rapidity y and the pseudo-rapidity η. They are defined as:

$$y = \frac{1}{2} ln\left(\frac{E + p_z}{E - p_z}\right) = \text{arctanh}\left(\frac{p_z}{E}\right) \tag{2.7}$$

$$\eta = -ln\left[tan\left(\frac{\theta}{2}\right)\right] = \text{arctanh}\left(\frac{p_z}{\mid \vec{p} \mid}\right) \tag{2.8}$$

where $\vec{p}$ is the three-momentum of the particle and p_z is its component along the z-axis. E represents the energy of the particle. y and η get values in the range $[-\infty, \infty]$, being zero for $\theta = \pi/2$ and $\pm\infty$ for $\theta = 0$ or $\theta = \pi$.
These latter two variables are introduced because the difference of rapidities between two particles is a Lorentz invariant quantity under boosts along the z-axis. Since Lorentz boosts are applied to move from the laboratory frame to the center-of-mass frame of the colliding particles, η and y are widely used instead of the polar angle θ. In case of an ultra-relativistic particle, the rapidity and the pseudo-rapidity are approximately equal, $\eta \approx y$, as shown in Eq. 2.7 and Eq. 2.8. The main difference between the two is that the rapidity depends both on the energy of the particle and its momentum, a combined measurement not trivial at high energies, while the pseudo-rapidity depends only on the particle's momentum. For these reasons, the pseudo-rapidity is preferred to the rapidity, making (r, η, ϕ) the used coordinate system.
The pseudo-rapidity and the azimuth angle, are used to define a geometrical distance, applied when computing the separation between two physics objects:

$$\Delta R = \sqrt{\Delta\eta^2 + \Delta\phi^2} \tag{2.9}$$

The third coordinate, r, is the radius from the center of the system. Actually, in this thesis, we use another information: the transverse momentum p_T. It is a 2D vector defined as the projection of the momentum in the transverse plane of the interaction. This parameter is related to the radius and it is important because it brings information about energy. In figure **??** it is possible to observe how p_T is calculated.

The inner detector relies on a 2 T solenoidal magnetic field, parallel to the beam axis, to reconstruct the tracks of charged particles. This information is used to measure the momentum of them using the sagitta method and the relation between momentum and radius of curvature, presented in (**??**).
The electromagnetic calorimeter [31] measures the energy of electrons, photons and hadrons. Once a particle among them hits the detector, an electromagnetic shower is generated and the energy of the particle is transferred to the material through radiation, as shown in figure **??**. Passive components of the calorimeter such as lead induce the loss of energy and the active ones, made by Liquid Argon (LAr) kept at a temperature of -184 °C, measure it. The shower is fully contained in this part of the detector, which is longer then 20 radiation lengths. For this reason this technique to measure energy is defined *destructive*.

The hadronic calorimeter has been placed to measure the energy of hadrons that survive to the electromagnetic calorimeter. It relies on the principle of particle shower, which is converted into an electric current and measured.
The muon spectrometer is the last component of the ATLAS experiment and measures the momenta of muons. In fact, they are similar to electrons, but with a greater mass, which leads only to weak interactions and more depth of penetration. This is why they are not revealed by inner systems.

The coordinate system adopted at ATLAS to describe events is discussed in detail in section **??**.

Chapter 3

Anomaly Detection

3.1 The LHC

Chapter 4

A particle physics analysis

4.1 The LHC

Chapter 5

ADJJ analysis

5.1 The LHC

Chapter 6

Conclusions

6.1 The LHC

Bibliography

[1] ATLAS Collaboration. *"Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC"*. *Phys. Lett. B*, 716:1–29, (2012).

[2] CMS Collaboration. *"Observation of a New Boson at a Mass of 125 GeV with the CMS Experiment at the LHC"*. *Phys. Lett. B*, 716:30–61, (2012).

[3] K. Garrett et al. *"Dark Matter: A Primer"*. *Advances in Astronomy*, (2011). doi: 10.1155/2011/968283.

[4] M. E. Peskin and D. V. Schroeder. *"An Introduction to quantum field theory"*. (1995). doi: 10.1201/9780429503559.

[5] Wikipedia The Free Encyclopedia. *"Standard Model"*, (2026). URL `https://en.wikipedia.org/w/index.php?title=Standard_Model&oldid=1368397978`.

[6] R. P. Feynman. *"A New Approach to Quantum Theory"*. Ed. by Brown Laurie M. World Scientific Publishing Co. Pte. Ltd., (2005).

[7] A. Pich. *"The Standard Model of Electroweak Interactions"*, (2012). URL `https://arxiv.org/abs/1201.0537`.

[8] E. Noether. *"Invariant variation problems"*. *Transport Theory and Statistical Physics*, (1971). URL `http://dx.doi.org/10.1080/00411457108231446`.

[9] W. Greiner et al. *"Quantum Chromodynamics"*. (2002). URL `https://doi.org/10.1007/978-3-662-04707-1`.

[10] S. Weinberg. *"A Model of Leptons"*. (1967). URL `https://link.aps.org/doi/10.1103/PhysRevLett.19.1264`.

[11] F. Takahashi et al. Particle Data Group. *"Review of Particle Physics"*. Table 11, (2026). URL `https://pdg.lbl.gov/2026/web/viewer.html?file=../reviews/rpp2026-rev-higgs-boson.pdf`.

[12] M. K. Gaillard J. Ellis and D. V. Nanopoulos. *"A Historical Profile of the Higgs Boson"*. (2016). URL `https://cds.cern.ch/record/2012465`.

[13] ATLAS Collaboration. *"Search for heavy neutral Higgs bosons decaying into a top quark pair in 140 fb^{-1} of proton-proton collision data at $\sqrt{s}$= 13 TeV with the ATLAS detector"*. *Journal of High Energy Physics*, 13, (2024).

[14] A. Strumia M. Cirelli and J. Zupan. *"Dark Matter"*, (2025). URL https://arxiv.org/abs/2406.01705.

[15] E. Hubble and M. L. Humason. *"The Velocity-Distance Relation among Extra-Galactic Nebulae"*. (1931). URL https://ui.adsabs.harvard.edu/abs/1931ApJ....74...43H.

[16] R. D. Blandford T. G. Brainerd and I. Smail. *"Weak Gravitational Lensing by Galaxies"*. (1996).

[17] D. Clowe et al. *"A Direct Empirical Proof of the Existence of Dark Matter"*. (2006). URL http://dx.doi.org/10.1086/508162.

[18] D. Alves et al. *"Simplified Models for LHC New Physics Searches"*. *J. Phys. G*, (2012).

[19] R. Torre D. Pappadopulo, A. Thamm and A. Wulzer. *"Heavy vector triplets: bridging theory and data"*. *Journal of High Energy Physics*, (2014).

[20] G. et al. Aad. *"Anomaly detection search for new resonances decaying into a Higgs boson and a generic new particle X in hadronic final states using* $\sqrt{s} = 13$ TeV *pp collisions with the ATLAS detector"*. *Phys. Rev. D*, (2023). URL https://link.aps.org/doi/10.1103/PhysRevD.108.052009.

[21] A. Tumasyan et al. *"Search for new heavy resonances decaying to WW, WZ, ZZ, WH, or ZH boson pairs in the all-jets final state in proton-proton collisions at s=13TeV"*. *Physics Letters, Section B: Nuclear, Elementary Particle and High-Energy Physics*, (2023). doi: 10.1016/j.physletb.2023.137813.

[22] G. Albouy et al. *"Theory, phenomenology, and experimental avenues for dark showers: a Snowmass 2021 report"*. *The European Physical Journal C*, (12). URL http://dx.doi.org/10.1140/epjc/s10052-022-11048-8.

[23] G. et al. Aad. *"Search for resonant production of dark quarks in the dijet final state with the ATLAS detector"*. *Journal of High Energy Physics*, (2024). URL https://doi.org/10.1007/JHEP02(2024)128.

[24] Esma Mobs. *"The CERN accelerator complex in 2019"*. (2019). URL https://cds.cern.ch/record/2684277.

[25] Communications Education and CERN Outreach Group. *"LHC the guide FAQ"*. URL https://cds.cern.ch/record/2255762.

[26] L. Evans and P. Bryant. *"LHC Machine"*. *Journal of Instrumentation*, (2008). URL https://doi.org/10.1088/1748-0221/3/08/S08001.

[27] L. Rossi. *"The LHC superconducting magnets"*. *Conf. Proc. C*, (2003).

[28] O. Brüning et al. *"LHC Design Report"*. CERN Yellow Reports: Monographs. (2004). URL https://cds.cern.ch/record/782076.

[29] ATLAS Collaboration. *"Cumulative integrated luminosity delivered to and recorded by the ATLAS Experiment during LHC Run 1, 2 and 3"*. (2026). URL `https://cds.cern.ch/record/2964765`.

[30] ATLAS collaboration. The ATLAS Experiment. URL: `https://atlas.cern/`.

[31] Riccardo Maria Bianchi and ATLAS Collaboration. *"ATLAS experiment schematic or layout illustration"*. (2022). URL `https://cds.cern.ch/record/2837191`.